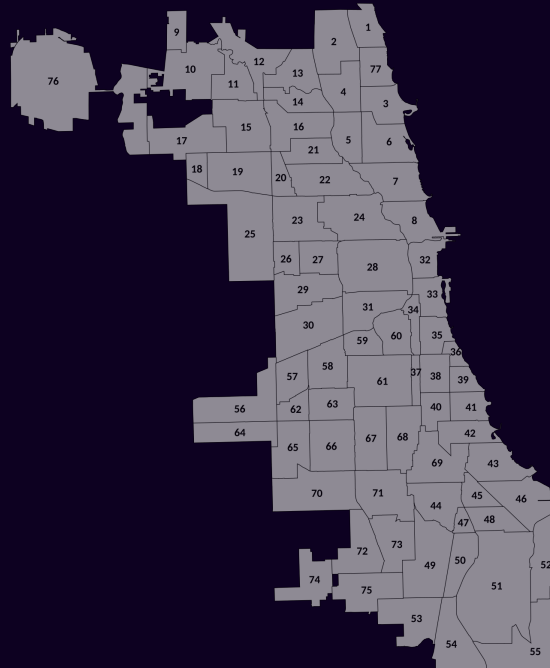


Adapting a Topological Framework from Biology to Compositional Crime Dynamics

Adapting a Mapper-based framework from microbiome dynamics to Chicago crime data



Research Question

Goal: Explore whether TDA can reveal structural changes in Chicago crime composition before, during, and after COVID-19.

Did Chicago community areas return to their pre-COVID crime composition, or did some transition into new persistent compositional regimes?

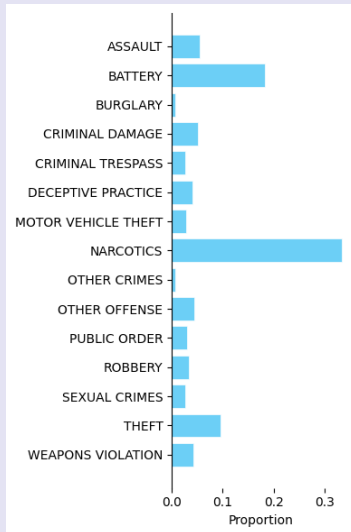
Composition (what type) vs. **Volume** (how much)

A community can record 200 incidents in different time periods. Example: Community A, pre-covid is dominated by Narcotics and post-covid by battery and narcotics, additionally other types non-significant increased their relative frequency.

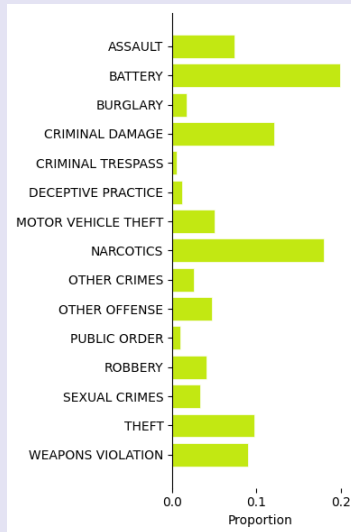
Same community different profiles

Community A (month)

Pre-Covid



Post-Covid



Total number of incidents might increase, decrease or remain. But that does not study what is happening with the type of crimes

Methodological Motivation Chang et al. (2021)

Chang et al. used Mapper to characterize microbiome dynamics. Mapper is used to represent high-dimensional compositional data as a landscape, where states and transitions can be interpreted through the topology of the graph.

Chang et al.

1. Samples: gut microbiome vectors (OTU abundances)
2. Jensen-Shannon distance matrix
3. Mapper with PCoA + rank as lens
4. k-NN density estimation on nodes ($k = 10\%$ samples)
5. Local maxima $\rightarrow$ to color states
6. Directed graph for state assignment
7. Interpretation

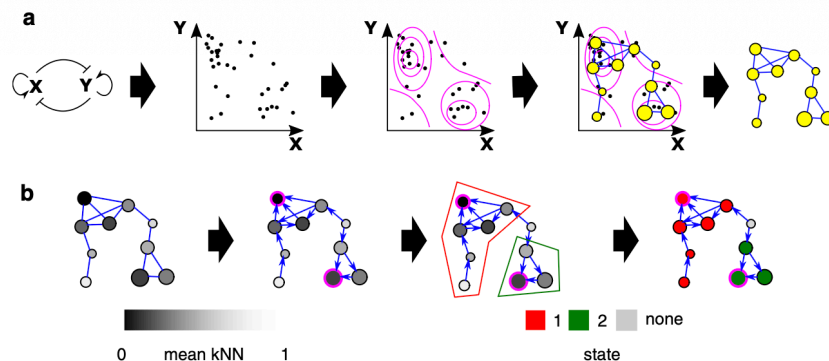


Fig 1. Chang et al.

The motivation is not to force crime data into clean clusters, but to ask whether TDA can reveal a meaningful temporal organization in a noisy social dataset.

Data

Chicago crime records

Chicago Data Portal · 2018–2023 · monthly by community area (77)

Periods

2018-2019: Pre-Covid (Baseline) | **2020-2021**: Covid (Disruption) | **2022 - 2023**: Post-Covid (Recovery).

77 community areas

Chicago's official statistical/geographic divisions.

Aggregation level

Community-month: raw counts of different crime categories

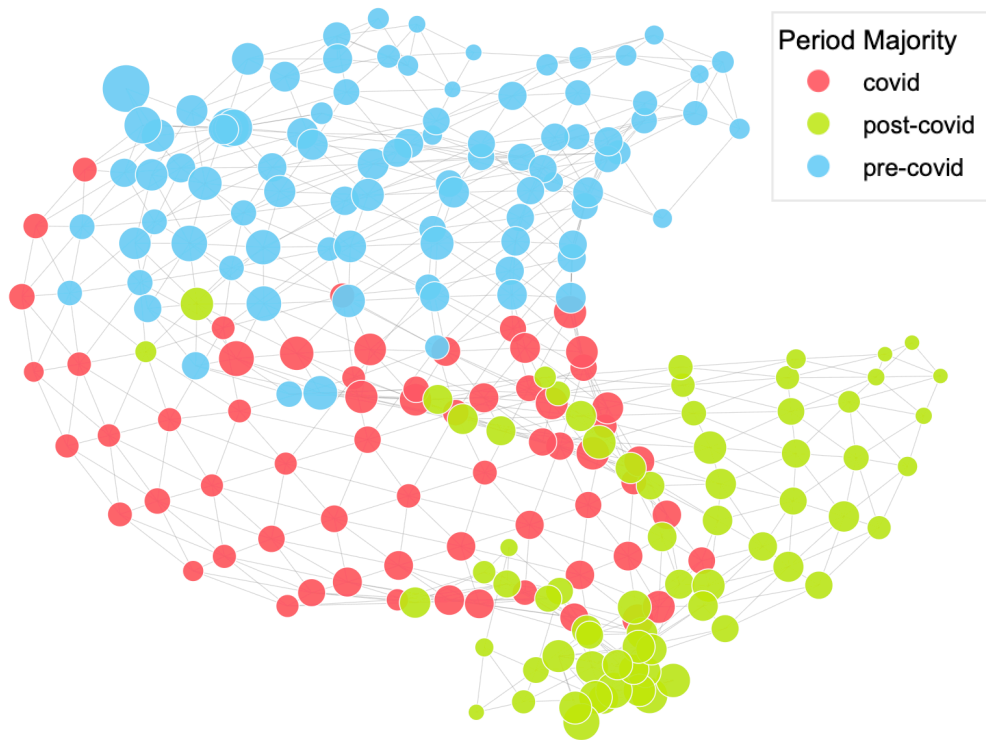
Supplementary (interpretation only)

ACS (American Community Survey): poverty rate, median income, unemployment, population composition, education level, etc.
Crime context: arrest rate, domestic violence rate.

Pipeline

0	Build compositional vectors	Modify raw counts (make samples comparable across communities & months: 200 theft, 100 battery, 50 burglary —> 57% theft, 29% battery, 14%burglary
1	Compute Jensen Shannon distance matrix	Pairwise compositional distances computed between community-month samples. JS distance compare samples by how similar or different their crime “distribution” was.
2	PCoA rank lens	Project distance matrix via PCoA, rank-transform for global ordering.
3	Apply Mapper (single linkage: JSD matrix)	Nodes (N) = overlapping clusters, edges = shared samples (min=1) Parameters where chose considering: $ N \leq 10\%$ total and minimizing $ N =1$
4	k-NN density estimation for each node	Estimate local density at each node using the inverse of the average k-nearest neighbors distance, with k=10% of the total
5	Identify local maxima = candidate states	Local maxima are nodes with higher estimated density than their neighbors.
6	Assign states to each node (directed graph) low density - - > high density	Each node is assigned to the closest local maximum through directed graph distance. Nodes equally close to multiple maxima are labeled as unstable or unassigned.
7	Interpret and validate.	

Mapper Graph (Temporal Structure)



Colors

Node color represents the majority period among the samples contained in that node.

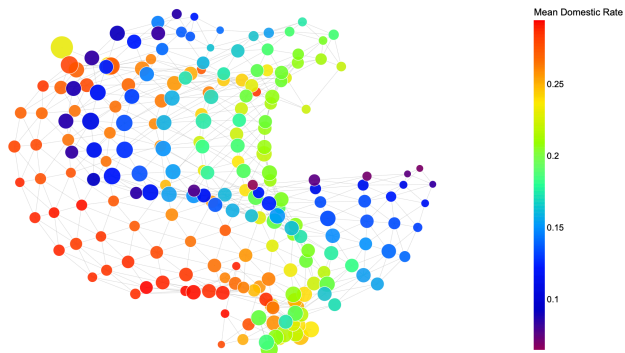
Observations

The graph is a single connected structure (connected compositional landscape), .

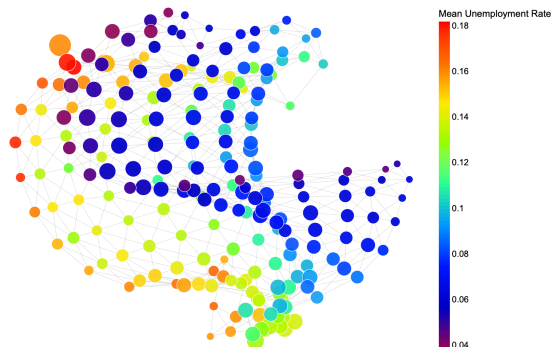
The crime composition remained continuous, but the periods seem to occupy different regions of the same landscape.

Region Profiles (ACS Features)

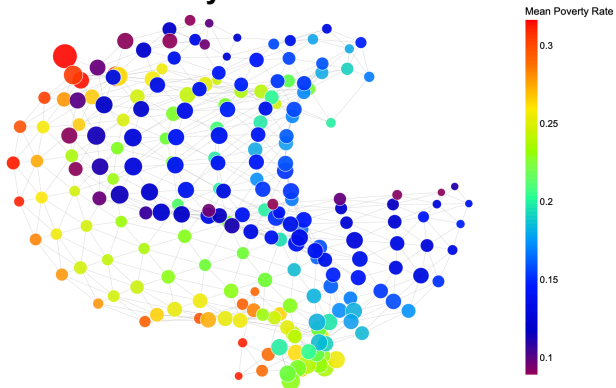
Domestic Violence Rate



Unemployment Rate



Poverty Rate



COVID-region profile

↑ Domestic incidents

↑ BATTERY share

↑ Poverty rate

↓ Median income

↓ Arrest rate

Observations (Note: ACS features used for interpretation only not included in Mapper inputs)

COVID-majority nodes occupy a recognizable region of the landscape. This region is associated with higher domestic violence rate, higher battery share, higher poverty, lower median income, and lower arrest rate.

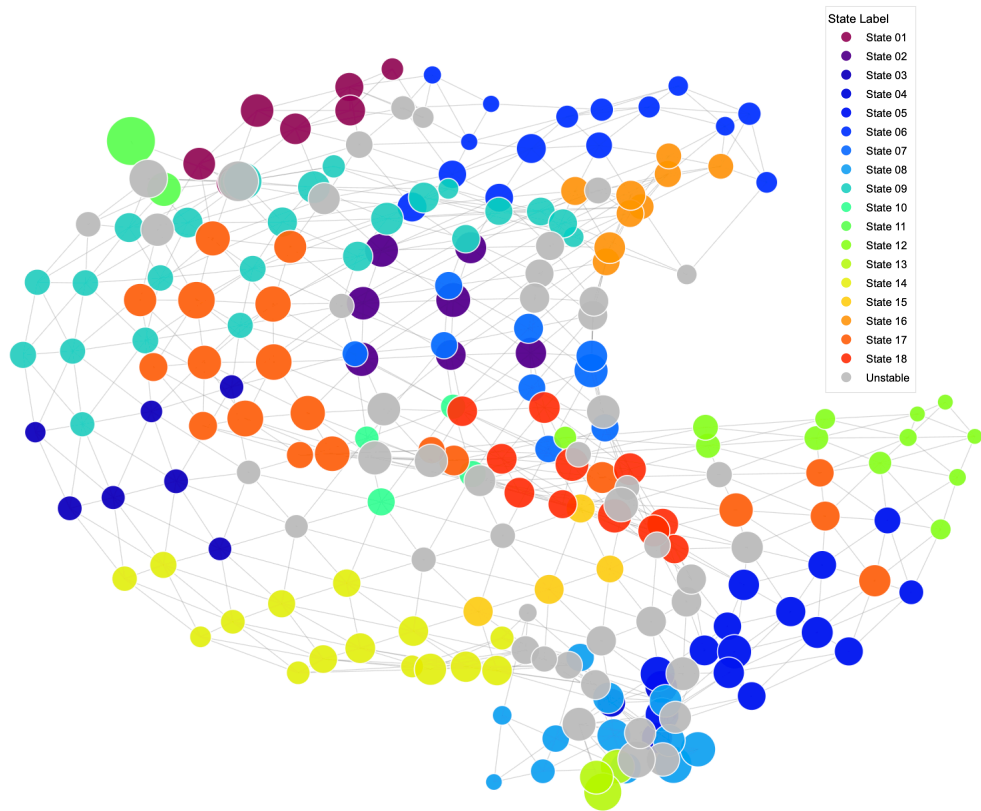
Important: The relation is not causal

Crime-Composition States (Local Maxima)

Observations

The local maxima give candidate states, but these states are not isolated clusters. They are density peaks within a connected graph. In my data, those peaks exist, but the boundaries are less [clear](#) and there is more instability between them, so I interpret them as recurring compositional neighborhoods instead of hard regimes.

Result: 18 “states” + 1 Unstable “state” (46 nodes)

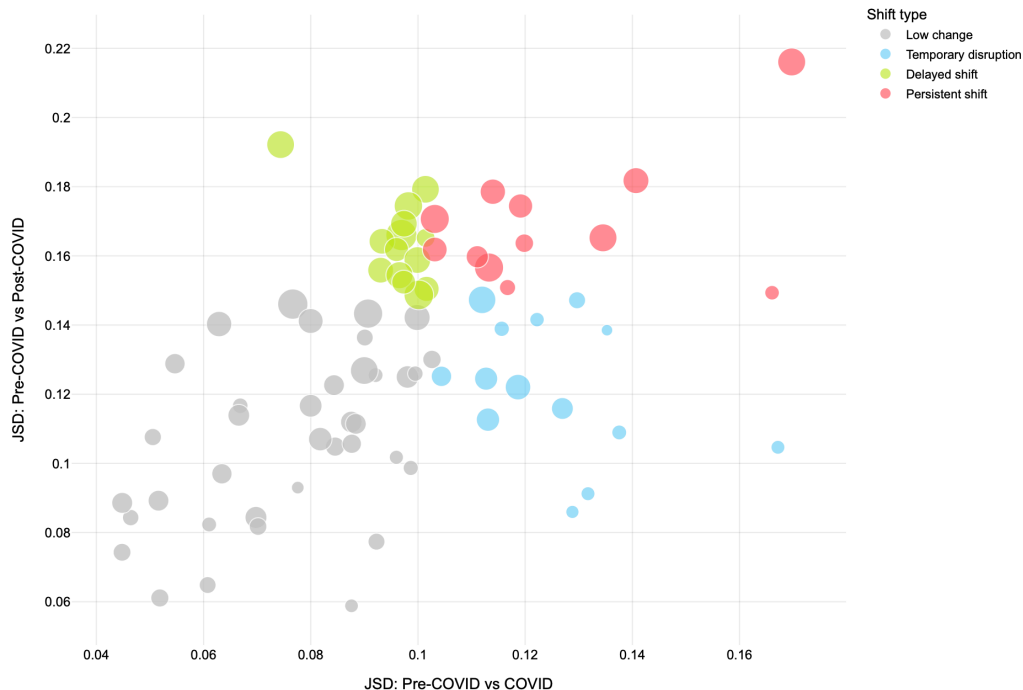


Gradual reorganization within a connected landscape, rather than a clean movement between separated regimes.

Community Shifts

Each community's shift type is determined by comparing its mean crime-share vector between the pre-COVID and COVID, and between the pre-COVID and post-COVID using (JSD)

The citywide pattern is not a universal transition. Most communities show low change or temporary disruption.



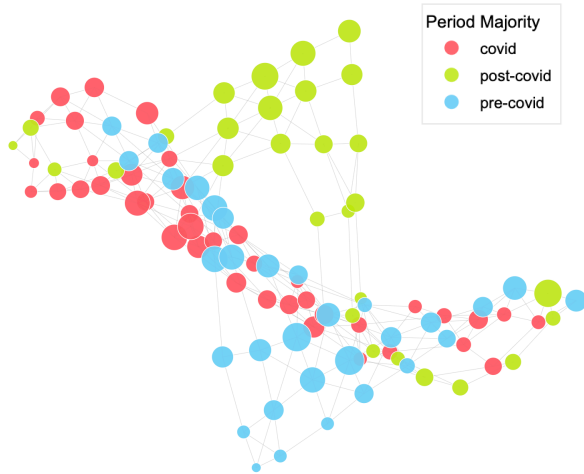
Some communities shifted more than others, motivating a closer look at spatial and socioeconomic context.

Shift type (manual)	# Communities
Low change	37
Temporary disruption	14
Delayed shift	14
Persistent shift	12
26 of 77 communities (34%) show stronger evidence of compositional shift under the JSD-based classification.	

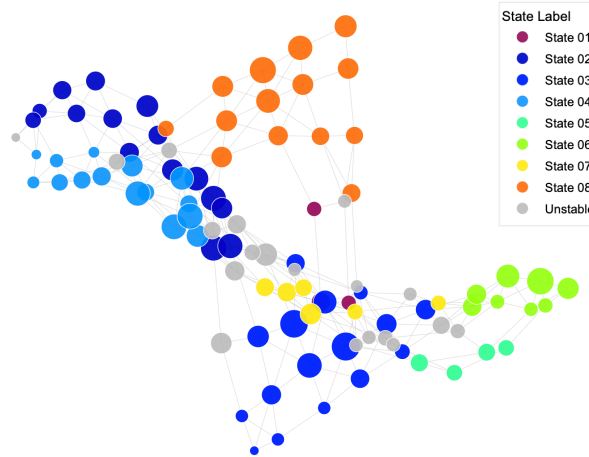
26 Communities Graph

To explore whether these 26 communities form a clearer landscape, I applied the same methodology only to samples from those communities. This was an exploratory subset analysis, not the main evidence. The results are here:

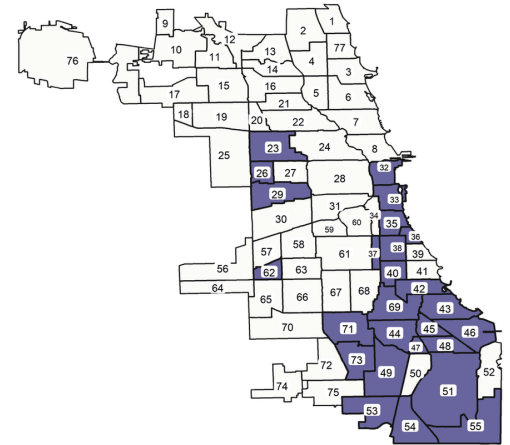
Mapper color by Period



Mapper color by States



Chicago Map divided by community



AI generated image

Some areas are geographically close.

Conclusions

The goal was not to find isolated clusters, but to use Mapper as a landscape representation of crime composition.

The resulting graph is continuous and densely connected. The detected “states” are better interpreted as density-based neighborhoods that summarize recurring crime composition profiles, not clean compositional modes.

Mapper preserved something interesting: the path. Two communities can show the same compositional shift and yet have traveled through different regions of the landscape to get there, that trajectory is invisible to direct compositional distance alone.

COVID disruption did not produce one citywide transition into a new discrete state. Still, the landscape shows temporal organization: periods occupy different regions, some states are more common during COVID or post-COVID, and some communities show stronger pre vs. post compositional shifts.

Why the result was not “clean” and the research question cannot be fully answered ?

Urban crime may change gradually by nature

Broad crime categories reduce compositional resolution (fine-grained data is more suitable)

Reporting patterns may have shifted during COVID

The framework was designed for data with strong, well-separated states. When the signal is weak, the setup preserves connectivity rather than revealing separation.

Appendix

JSD - Jensen Shannon Distance

Measures the similarity between two probability distributions.

Kullback-Leibler Divergence (not a metric)

$$D_{\text{KL}}(P \parallel Q) = \sum_{x \in \mathcal{X}} P(x) \log \frac{P(x)}{Q(x)}.$$

How much an approximating probability distribution Q is different from a true probability distribution P

JSD is a symmetrized and smoothed version of KL:

$$\text{JSD}(P \parallel Q) = \sqrt{\frac{1}{2} D_{\text{KL}}(P \parallel M) + \frac{1}{2} D_{\text{KL}}(Q \parallel M)},$$

$$\text{With } M = \frac{1}{2}(P + Q)$$

Density (KNN-distance k=10% samples)

KNN distance of a point "i" is defined:

$$\text{kNN}(i, k) = \frac{\sum_{j \in N(k)_i} d_{ij}}{k}$$

where d_{ij} is the distance between points i and j .

For a vertex V representing n points:

$$D_{\text{inv}}(V) = \frac{\sum_{i \in V} \text{kNN}(i, k)}{n^2}$$

Finally, the density of a vertex/node is defined as:

$$D(V) = \frac{1}{D_{\text{inv}}}$$

Appendix (Expectation)

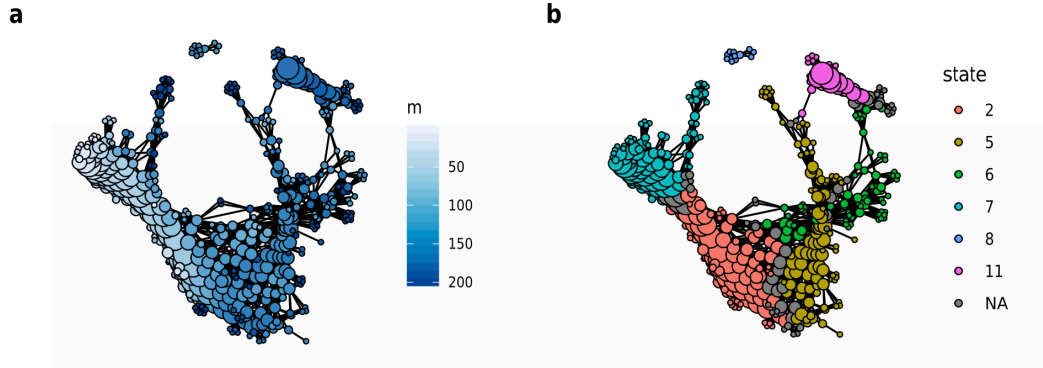


Fig. 5 (Chang et al.) The combined phase space of two *Prochlorococcus* communities inhabiting the Atlantic and Pacific Oceans.

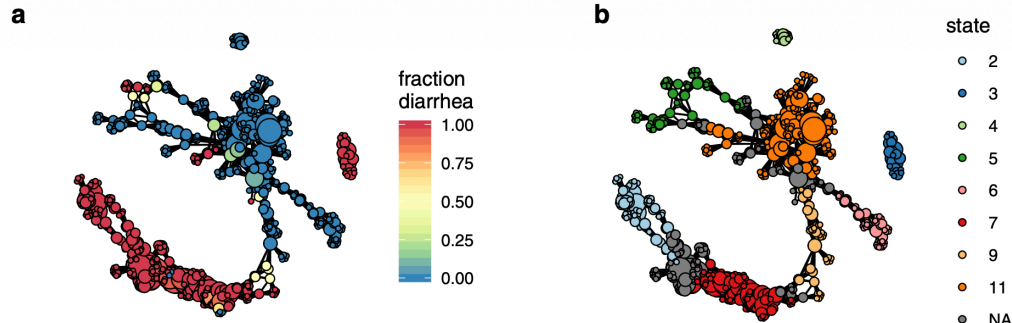
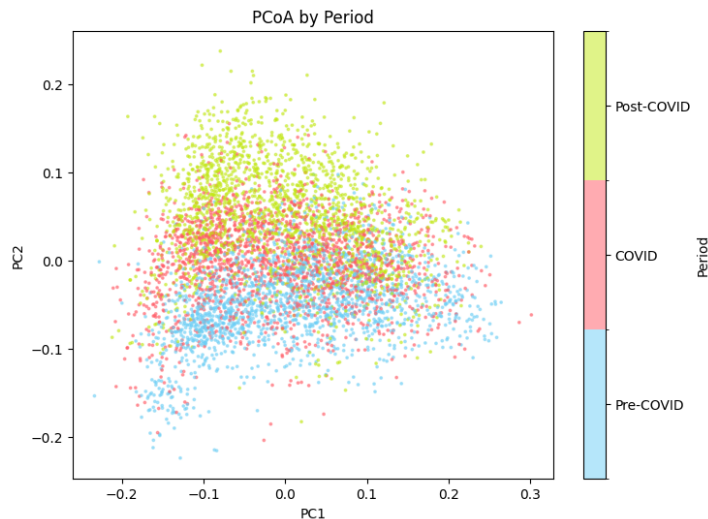


Fig. 2 (Chang et al.) The phase space of the cholera gut microbiome.

Point Cloud



The PCoA coordinates were ranked before constructing the Mapper cover. This keeps the relative ordering of samples along each PCoA axis, but reduces the influence of outliers and uneven spacing.

PCoA RANK

Observations

The selected lens is useful for detecting gradual temporal organization, but less suited for discovering sharply separated regimes if those regimes are weak.

